

ECE462L PWM, ADC, and PCB Design Report

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Contents

1	Pulse-Width Modulation: Lab 1, Week 1	3
1.1	Introduction and Lab Purpose	3
1.2	Leaning, Success, and Struggles	6
2	Analog-to-Digital Conversion: Lab 1, Week 2	6
2.1	Introduction and Lab Purpose	6
2.2	Voltage Watch Window Screenshots	7
2.3	Voltage Calculation Code	7
2.4	Learning, Success, Struggles	8
3	Printed Circuit Board Layout: Lab 2	9
3.1	Schematic	9
3.1.1	General layout and flow of the board	9
3.1.2	Top layer	10
3.1.3	Ground layer (upper inner layer)	11
3.1.4	Power layer (lower inner layer)	12
3.1.5	Bottom layer	13
3.2	Loop Inductance	13
3.2.1	Importance of the critical and gating loops	13
3.2.2	Critical loop	14
3.2.3	High-side loop	15
3.2.4	Low-side loop	16
3.3	Node Capacitance	16
3.3.1	Importance of switch node capacitance	16
3.4	Sensitive Signals	18
3.5	3D Layout	19
3.6	Learning, Success, Struggles	19
4	Conclusion	20
4.1	Highs and Lows	20
4.2	Future Goals	20

1 Pulse-Width Modulation: Lab 1, Week 1

1.1 Introduction and Lab Purpose

The purpose of the lab is to configure ePWM Module with a fixed duty cycle using the TI C2000 Microcontroller. This ePWM Module generates ePWM logic signals as switching pulses, which are then fed to a Gate Driver. The Gate Driver will convert these weak logic pulses into a stronger waveform that modulates the power stage of the MOSFETs, turning them on and off.

More specifically, the ePWM module will generate two complementary switching pulses, one with duty D and one with duty D', to drive two MOSFETs in a half-bridge configuration. There will be a tiny delay between these two complementary switching pulses, called a "dead time," to account for the time that both devices are off. As a result, this dead time allows one transistor to fully transition from OFF to ON after the other fully transitions from ON to OFF, and vice versa. This will avoid the case where both transistors are ON, causing a destructive short circuit that damages the half-bridge.

Table 1: Register/bit values for each PWM setup

Register/Bit	PWM 1	PWM 2	PWM 3
Switching Frequency (f_{sw})	200 kHz	75 kHz	10 kHz
Counter Type	Up-Down Count	Down Count	Up Count
Duty	0.5	0.7	0.3
EPwm1Regs.ETSEL.bit.SOCAEN	0	0	0
EPwm1Regs.ETSEL.bit.SOCASEL	4	4	4
EPwm1Regs.ETPS.bit.SOCAPRD	1	1	1
EPwm1Regs.TBCTL.bit.CLKDIV	0b00	0b00	0b00
EPwm1Regs.TBCTL.bit.HSPCLKDIV	0b00	0b00	0b00
EPwm1Regs.TBCTL.bit.CTRMODE	0b10	0b01	0b00
EPwm1Regs.TBPRD	375	2000	15000
EPwm1Regs.CMPA.bit.CMPA	188	1400	4500
EPwm1Regs.AQCTLA.bit.CAD	0b10	0b10	0b00
EPwm1Regs.AQCTLA.bit.CAU	0b01	0b00	0b01
EPwm1Regs.AQCTLB.bit.CAD	0b01	0b01	0b00
EPwm1Regs.AQCTLB.bit.CAU	0b10	0b00	0b10
EPwm1Regs.AQCTLB.bit.PRD	0b00	0b01	0b00
EPwm1Regs.AQCTLB.bit.ZRO	0b00	0b00	0b10
EPwm1Regs.AQCTLA.bit.PRD	0b00	0b10	0b00
EPwm1Regs.AQCTLA.bit.ZRO	0b00	0b00	0b01
GpioCtrlRegs.GPAMUX1.bit.GPIO0	0b01	0b01	0b01
GpioCtrlRegs.GPAGMUX1.bit.GPIO0	0b00	0b00	0b00
GpioCtrlRegs.GPAMUX1.bit.GPIO1	0b01	0b01	0b01
GpioCtrlRegs.GPAGMUX1.bit.GPIO1	0b00	0b00	0b00

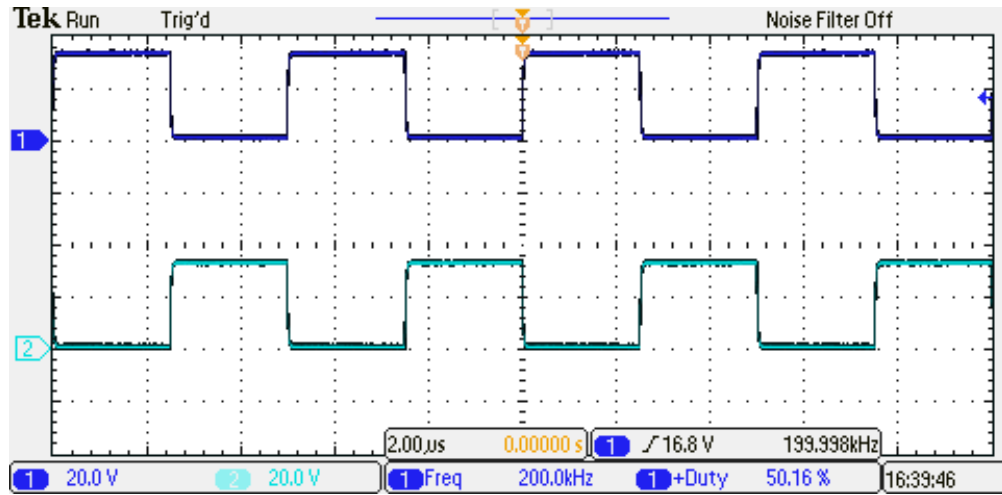


Figure 1: Measured PWM 1 D and D' signals

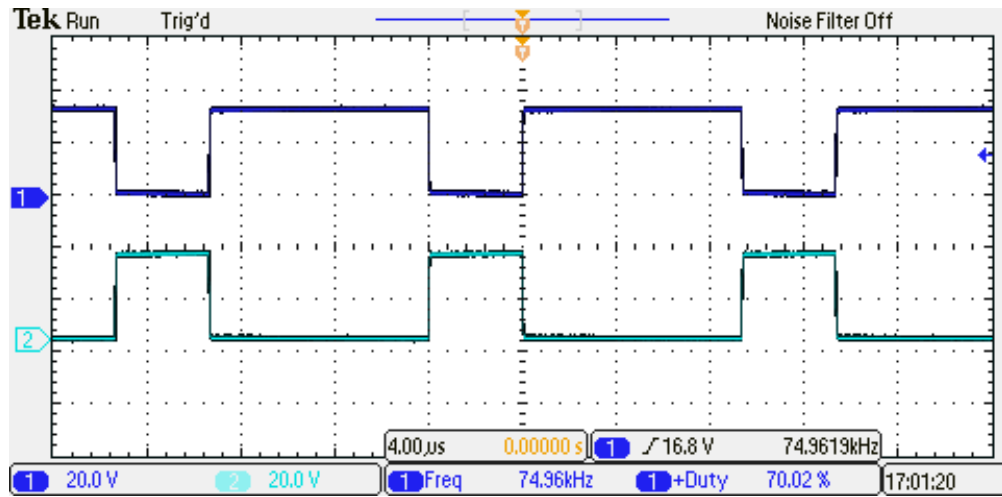


Figure 2: Measured PWM 2 D and D' signals

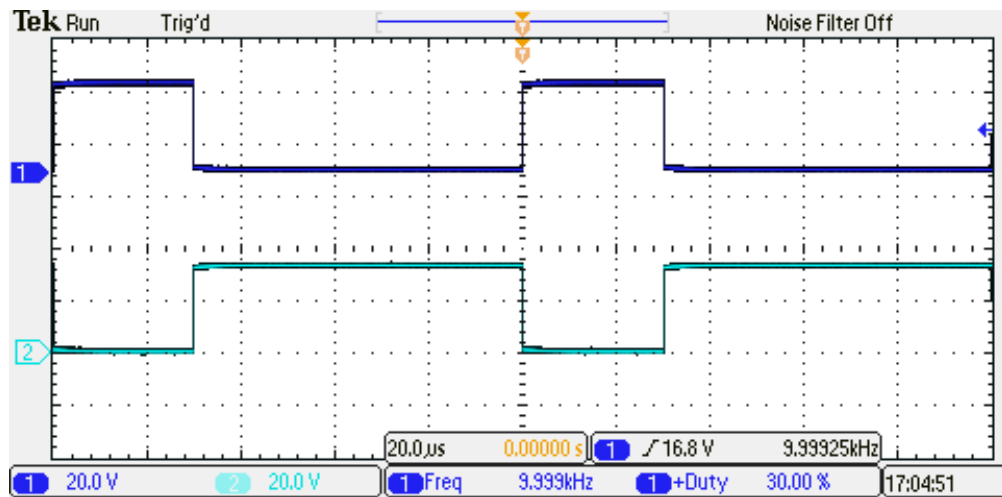


Figure 3: Measured PWM 3 D and D' signals

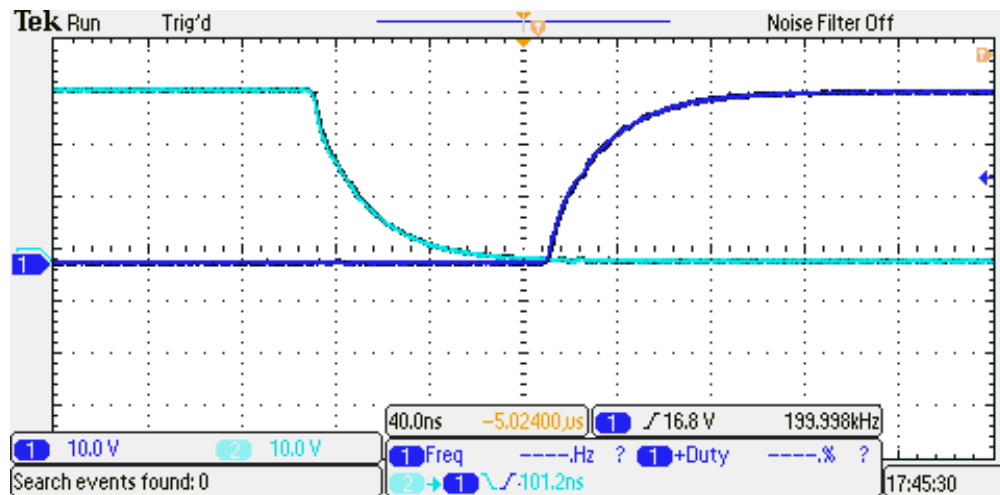


Figure 4: Rising edge dead-time

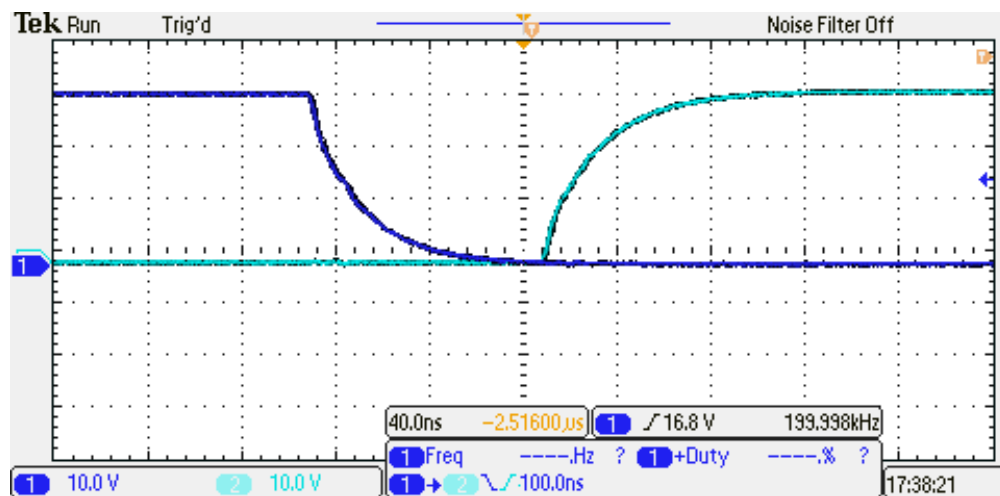


Figure 5: Falling edge dead-time

1.2 Leaning, Success, and Struggles

We struggled the most this lab with handling registers/bits properly since our group had no prior exposure to microcontroller programming past Lab 0. After some reading, we learned how to properly set bits and use binary in Code Composer. It also took a while to understand exactly how CMPA interacts with the counter to produce the duty cycle. However, we eventually succeeded with this, and were able to produce the correct output signals.

We primarily learned that the duty cycle is determined by the counter going above or below CMPA. Thus, we can raise CMPA to reduce the amount of time the counter spends “above” it, reducing the time the duty cycle is active. We also learned that we can change whether the duty is ON or OFF when the counter reaches the top or bottom. This allowed us to easily use the beginning/end of the up counter and down counters to get the desired duty cycle in combination with the CMPA value.

2 Analog-to-Digital Conversion: Lab 1, Week 2

2.1 Introduction and Lab Purpose

The purpose of this lab is to configure the ADC Module using a TI Microcontroller so that the timing between the ADC sampling and the periodic ePWM waveform is synchronized. More specifically, when the allowable voltage from the sensing circuit is applied to the delicate ADC pin on the Microcontroller, the ADC module will take an analog sample of the voltage at the ADC pin and convert it to a digitized value. The ADC module will be configured to use peak-valley sampling. This means that ADC SOC (Start of Conversion) will be triggered when the PWM counter reaches its maximum count value, “Nr” or zero. Meanwhile, the ADC EOC (End of Conversion) will be triggered when the PWM counter reaches zero or its maximum count value “Nr”, respectively.

Table 2: ADC Sampling Register/bit values

Register/Bit	Sampling Frequency		
	Peak-Valley	Once a cycle	Every five cycles
AdcaRegs.ADCSOC0CTL.bit.CHSEL	1	1	1
AdcaRegs.ADCSOC0CTL.bit.ACQPS	9	9	9
AdcaRegs.ADCSOC0CTL.bit.TRIGSEL	5	5	5
AdcaRegs.ADCINTSEL1N2.bit.INT1SEL	0	0	0
AdcaRegs.ADCINTSEL1N2.bit.INT1E	1	1	1
AdcaRegs.ADCINTFLGCLR.bit.ADCINT1	1	1	1
EPwm1Regs.ETSEL.bit.SOCASEL	3	3	3
EPwm1Regs.ETPS.bit.SOCAPRD	1	N/A	N/A
EPwm1Regs.ETPS.bit.SOCAPSSSEL	0	1	1
EPwm1Regs.ETSOCPS.bit.SOCAPRD2	N/A	2	10

2.2 Voltage Watch Window Screenshots

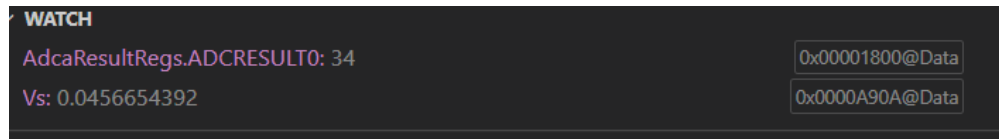


Figure 6: Measured Voltage = 0 Volt



Figure 7: Measured Voltage = 1 Volt



Figure 8: Measured Voltage = 2 Volts



Figure 9: Measured Voltage = 3 Volts



Figure 10: Measured Voltage = 4 Volts

2.3 Voltage Calculation Code

Listing 1: ADC Voltage Calculation

```
1  adcAResults[index++] = AdcaResultRegs.ADCRESULT0;  
2  Vs = (R1+R2)/R1 * AdcaResultRegs.ADCRESULT0 * Vfs /bit_range;
```

2.4 Learning, Success, Struggles

In this lab, we learned how to properly set up ADC calculations on the C2000. We quickly succeeded in learning how to properly trigger the ADC calculation process at the right events. However, we struggled for a while on how to modify the code to allow for sampling at different rates, such as only once per switching cycle or once every five switching cycles. After thoroughly reading and understanding each bit, we realized they allowed us to essentially delay the calculation until a certain number of "events" were triggered (SOCAPRD2), but we had to change other bits to allow SOCAPRD2 to be used (SOCAPSSEL). The difference between SOCAPRD2 and SOCAPRD, which both essentially have overlapping roles depending on the delay needed, is what confused us. We eventually did succeed in understanding it, however! Creating a table during the lab, similar to what is seen on the report, really bolstered our understanding.

3 Printed Circuit Board Layout: Lab 2

3.1 Schematic

3.1.1 General layout and flow of the board

The overall view of our board can be seen below in Fig. 11. The general layout of the board consists of the 24 V power input in the top-left, the 48 V power output in the bottom-left, the C2000 in the top-right, and the inductor in the bottom-right. The power flows in a clock-wise loop from the input through the fuse to the labeled gate driver in Fig. 11 and back out the 48 V jacks. There is clean space left between the gate driver and the 24 V input in order to give a clear path for the returning high $\frac{di}{dt}$ currents to return through. This will be explored in-depth later on.

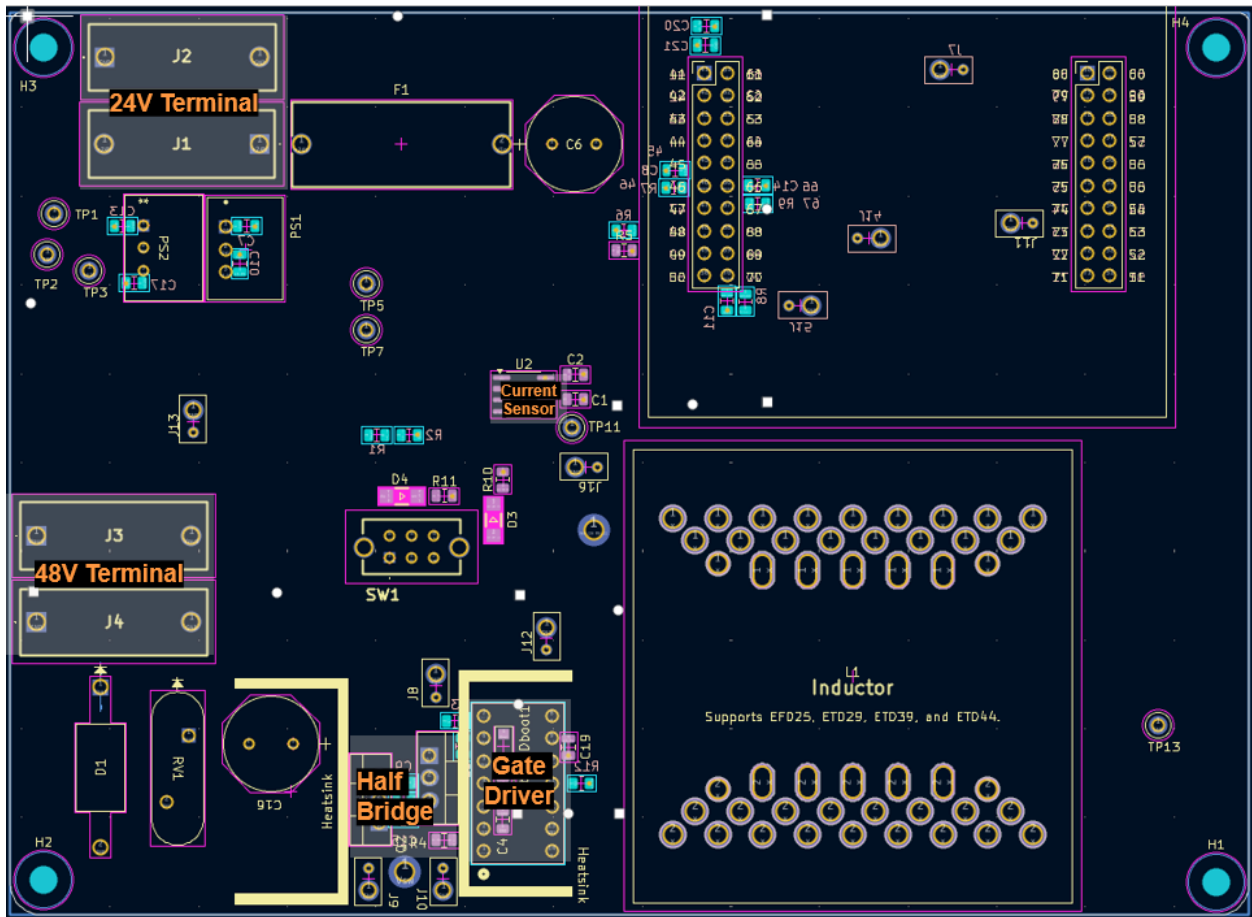


Figure 11: Overall Layout

3.1.2 Top layer

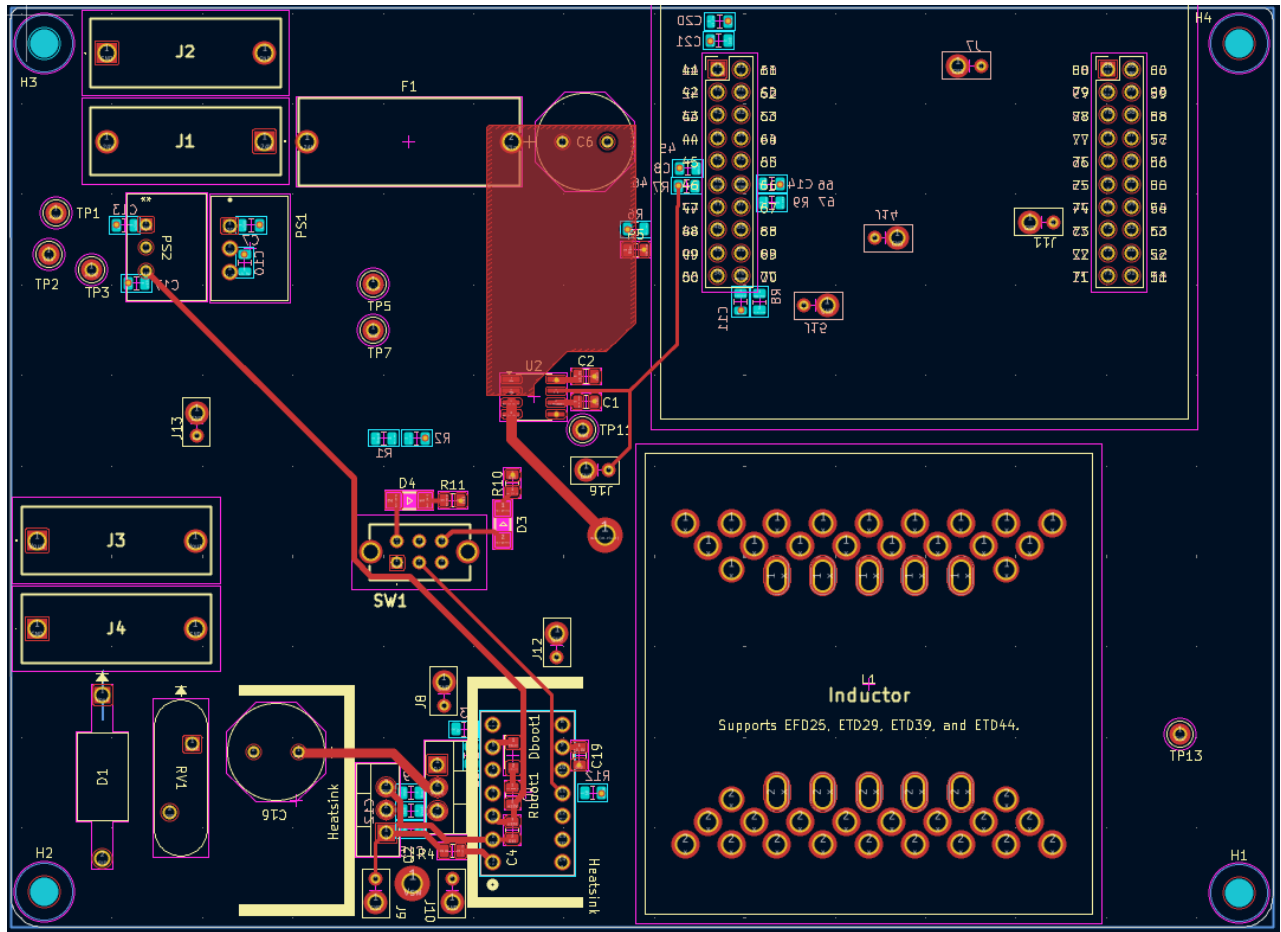


Figure 12: Top Layer

The top layer is one of our signal layers. The top layer connects the lower voltage (ex. 12 V paths) to the gate driver, and it also connects a few ADC routes. The top layer is mainly used to handle connections near the gate driver and critical loop. It also has a plane to help handle the input current into the current sensor.

The ground layer is pretty simple overall. It covers the entire board, and is accessed by vias. There are cutouts under the V_{sw} node in order to reduce capacitive effects due to rapidly changing voltages.

3.1.4 Power layer (lower inner layer)

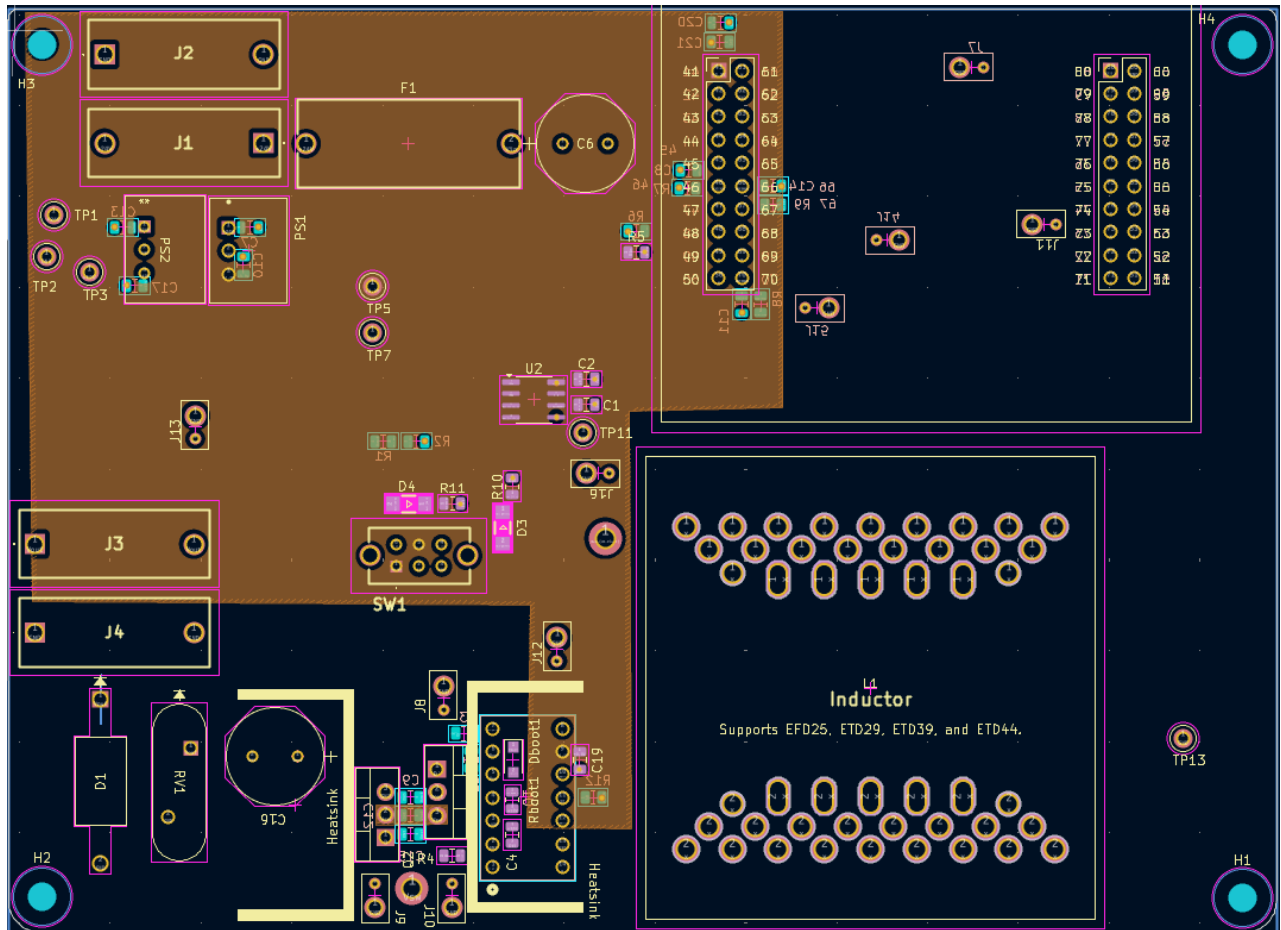


Figure 14: Power Layer

Our third layer, or second/lower inner layer, is the power layer. The majority of this layer is a large plane intended to distribute 3.3 V power to anywhere that requires it. Due to the vast space available in this layer, we also take advantage of this layer by using it in the critical loop to help keep it compact.

3.1.5 Bottom layer

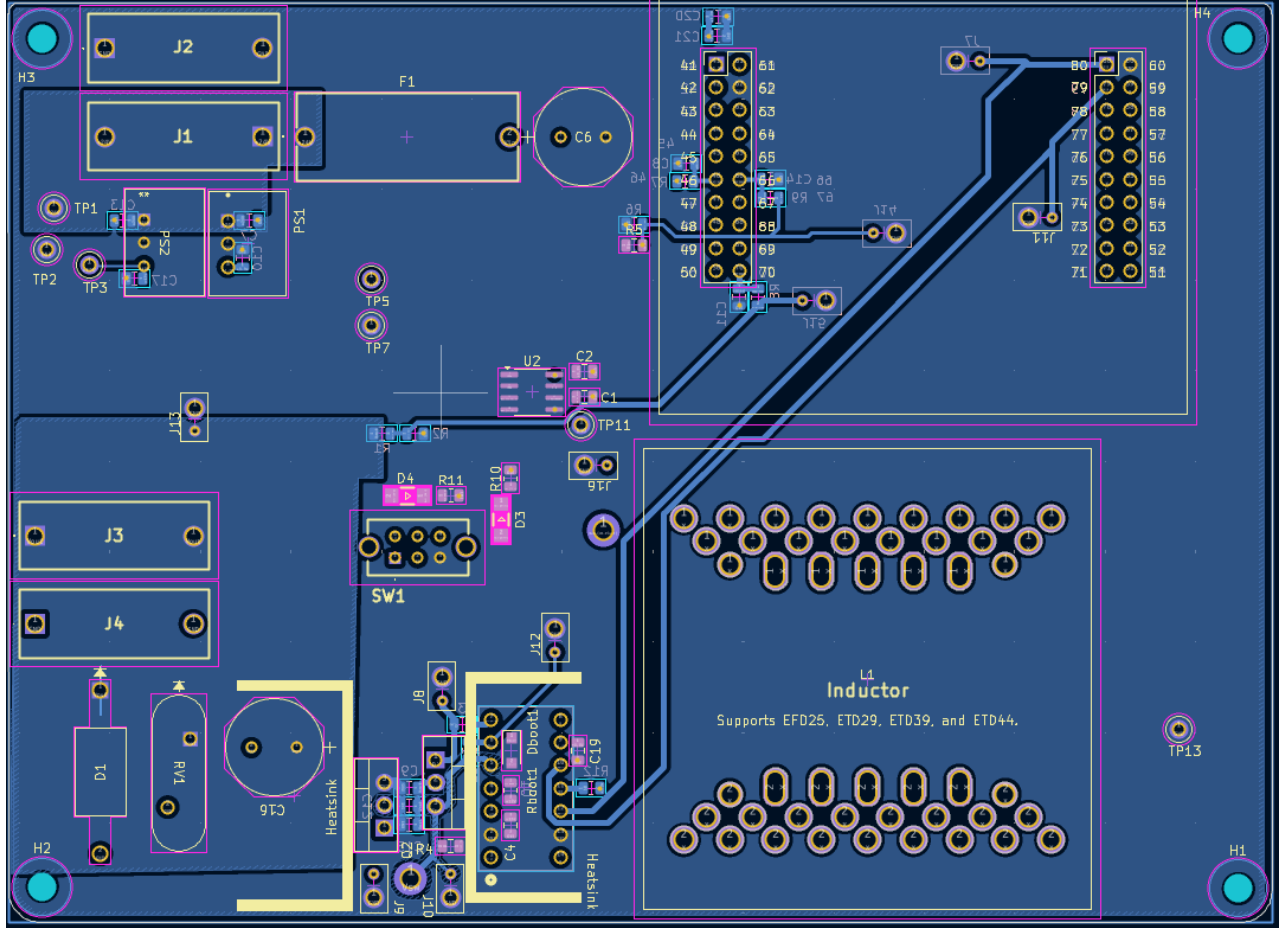


Figure 15: Bottom Layer

The bottom layer is a mix of signal and power. We use a plane on the bottom layer to help connect the input voltages (fuse, input jacks, and 12 V / 3.3 V buck converters) as well as the 48 V output nodes. We use large planes since there is a significant amount of current/power flowing. This layer is also used heavily in the critical loop as well as the ADC connections. There is also a large ground plane in other areas as the PCB manufacturer recommended it for copper balancing.

3.2 Loop Inductance

3.2.1 Importance of the critical and gating loops

Both the critical loop and the gating loop are crucial because every conductor that carries current will generate a magnetic field around it. As current flows through the conductor, forming the loop, inductance opposes a change in current by inducing a voltage, thereby storing energy in the magnetic field. The inductance of the loop is proportional to the surface area of the loop. As the loop becomes larger, it will contain more magnetic field, resulting in higher parasitic inductance.

Additionally, the Power MOSFETs, which operate at a very high switching frequency, reside in both the critical loop and the gating loop. At high switching frequency, the impedance from the high parasitic inductance will increase. As the sharp-edge switching current (high $\frac{di}{dt}$) goes through the high impedance from the parasitic inductance, large voltage spikes will be generated during the switching transition, which could damage the Power MOSFET devices.

3.2.2 Critical loop

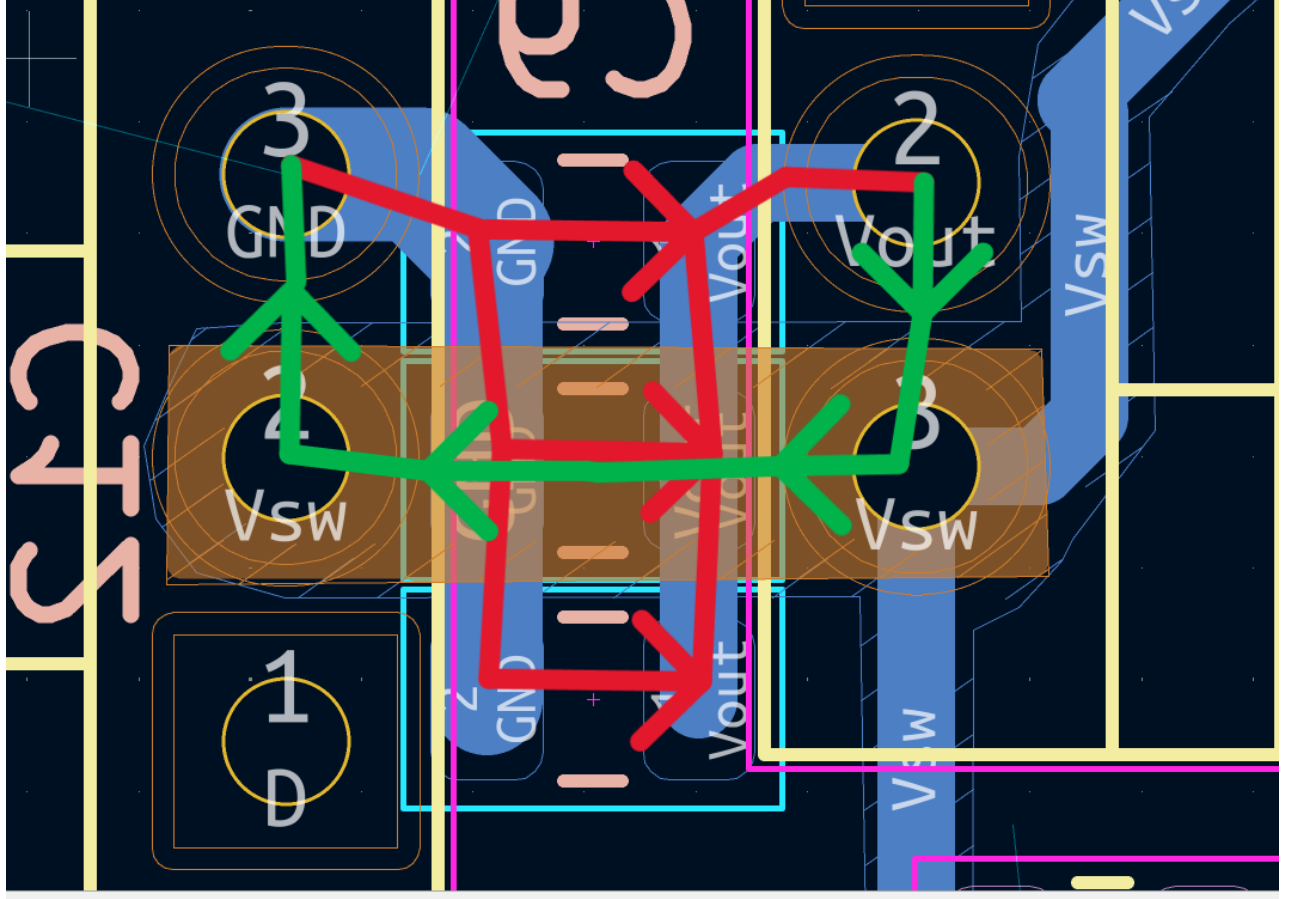


Figure 16: Critical Loop

To minimize the increase in the parasitic inductance generated in the critical loop, we placed each of the components in the loop close to each other, utilizing all four different layers of the PCB as seen in Fig. 16. We leveraged the fact that the inner layers are close to the outer layers, since most of the PCB's thickness is in the center. As a result, we stacked the bottom layer and a plane on the power layer to produce a very tight critical loop that maximizes the use of 3D space. The MOSFETs are also offset to make the switch node as tight as possible for the loop. The capacitors are packed as close as possible by placing them on the opposite side of the board from the MOSFETs.

3.2.3 High-side loop

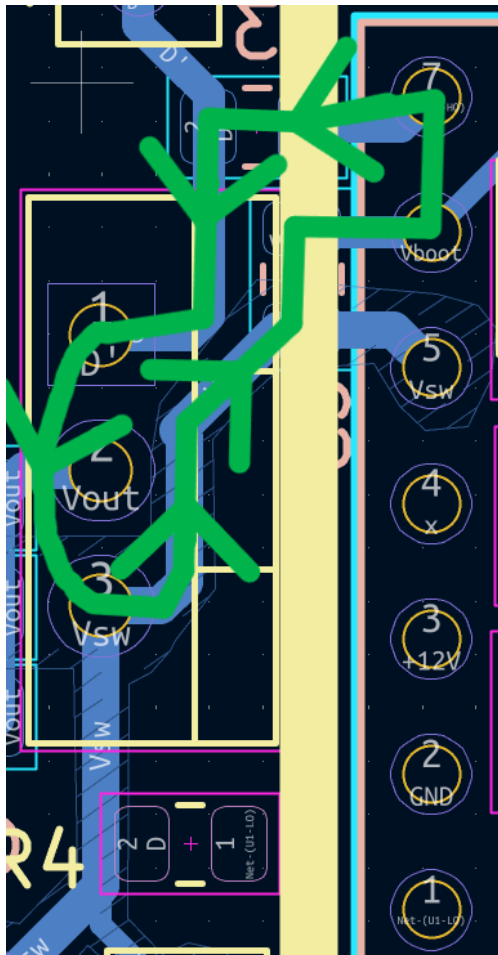


Figure 17: High-side charging loop

The high-side loop is shown above in Fig. 17. We minimized the high-side loop area by keeping the traces as close together as possible. For example, the V_{sw} node is connected directly upwards rather than diagonally to the gate driver IC to minimize the loop area. We also placed the resistor and capacitor on the path between the MOSFET and IC to further reduce the area. We considered flipping the parts to opposite sides of the board to further minimize the loop area, but doing so would have interfered with the heat sink and other components. We considered using more layers, but given the space between the pins on both the gate driver and MOSFET, we did not find the increase in area from the thickness of the board worthwhile.

3.2.4 Low-side loop

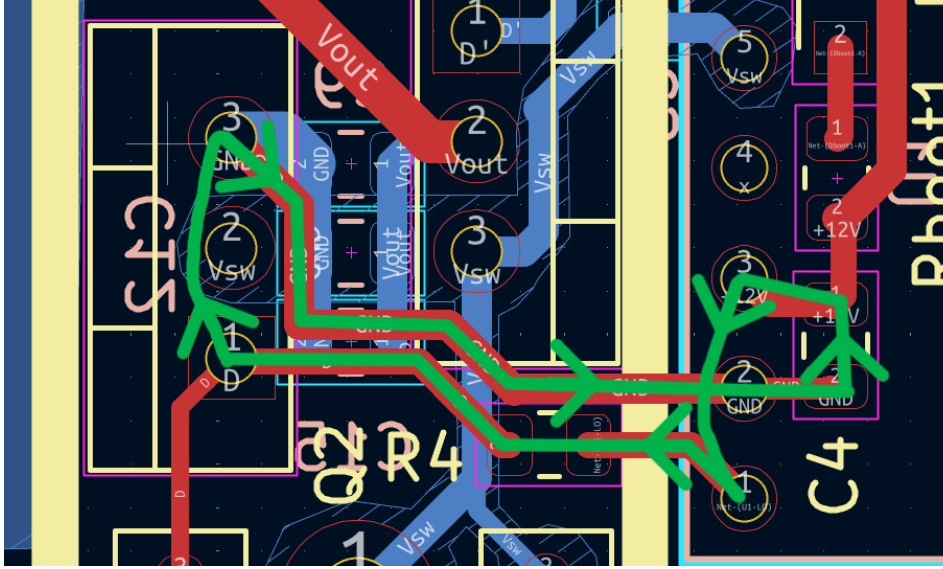


Figure 18: Low-side charging loop

The low-side loop seen above in Fig. 18 is minimized by keeping the traces as close together and as parallel as possible. We also placed some components on the opposite side of the gate driver to further pack components together. We considered moving the traces to another layer to lay them on top of each other. However, since the closest layer is ground, which makes routing difficult, we would have to choose a layer on the opposite half of the PCB. Since this would mean a large distance due to the thick center layer of the PCB, we did not find this option worthwhile since they are already so closely packed on the same layer.

3.3 Node Capacitance

3.3.1 Importance of switch node capacitance

Switch node capacitance is the parasitic capacitance created by high $\frac{dv}{dt}$ switching node from the inductor bobbin coupling with the ground plane. It is essential to minimize switch node capacitance because each MOSFET switching operation causes the switch node voltage to swing, resulting in switching losses from charging and discharging the switch node capacitance in addition to noise from the voltage/current leaking into the coupled planes.

To minimize switch node capacitance, we created a keep-out zone by carving out the ground plane above the switch node traces as seen in Fig. 19, thereby eliminating parasitic capacitance between the ground plane and the traces. This is acceptable in this case because we have a separate ground return path as seen in the low-side loop in Fig. 18. Note that part of these loops still cross the switch node slightly. We accepted this trade-off because we prioritized reducing the loop area to reduce parasitic inductance, and the area crossing the switch node is not too large.

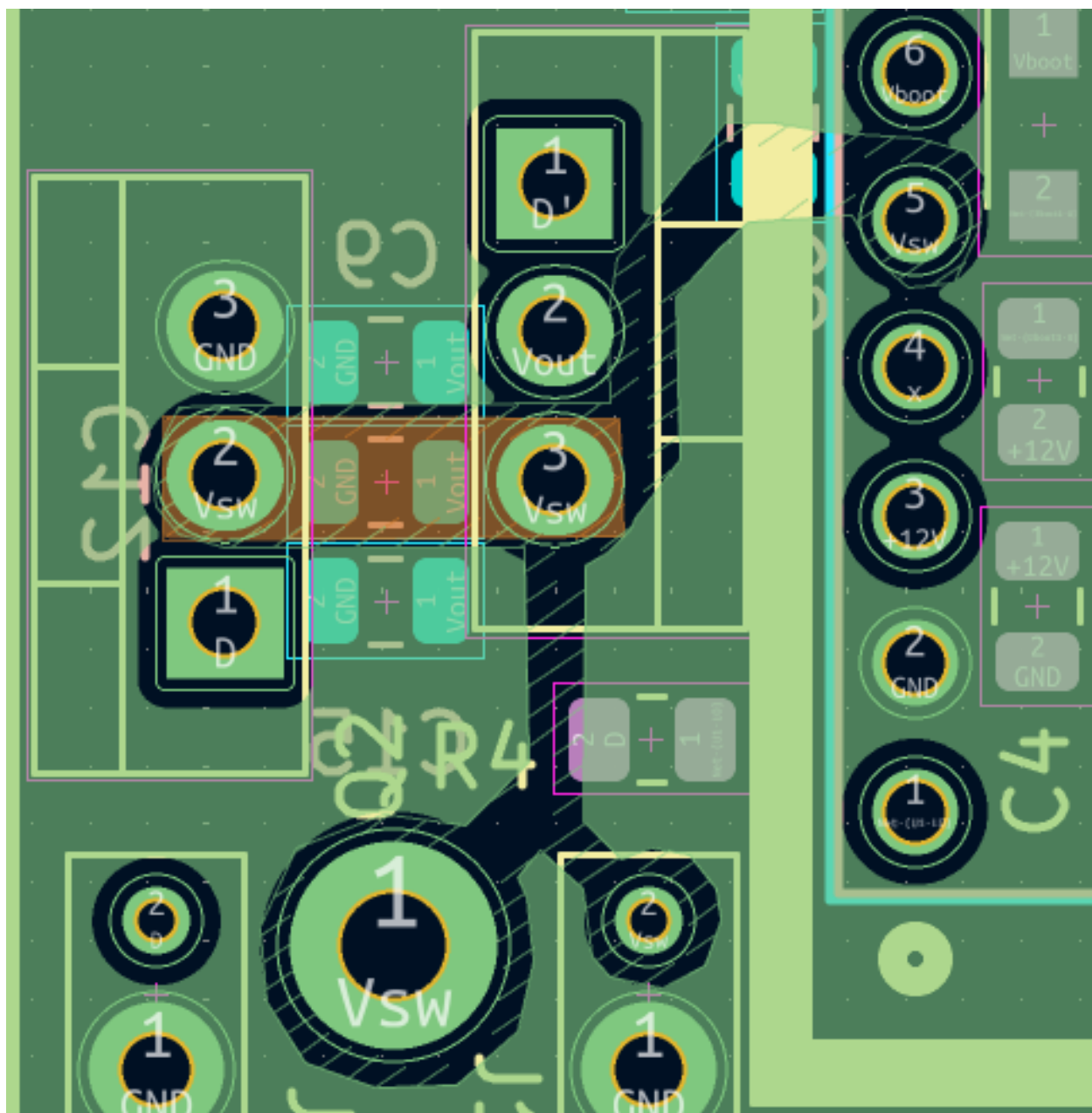
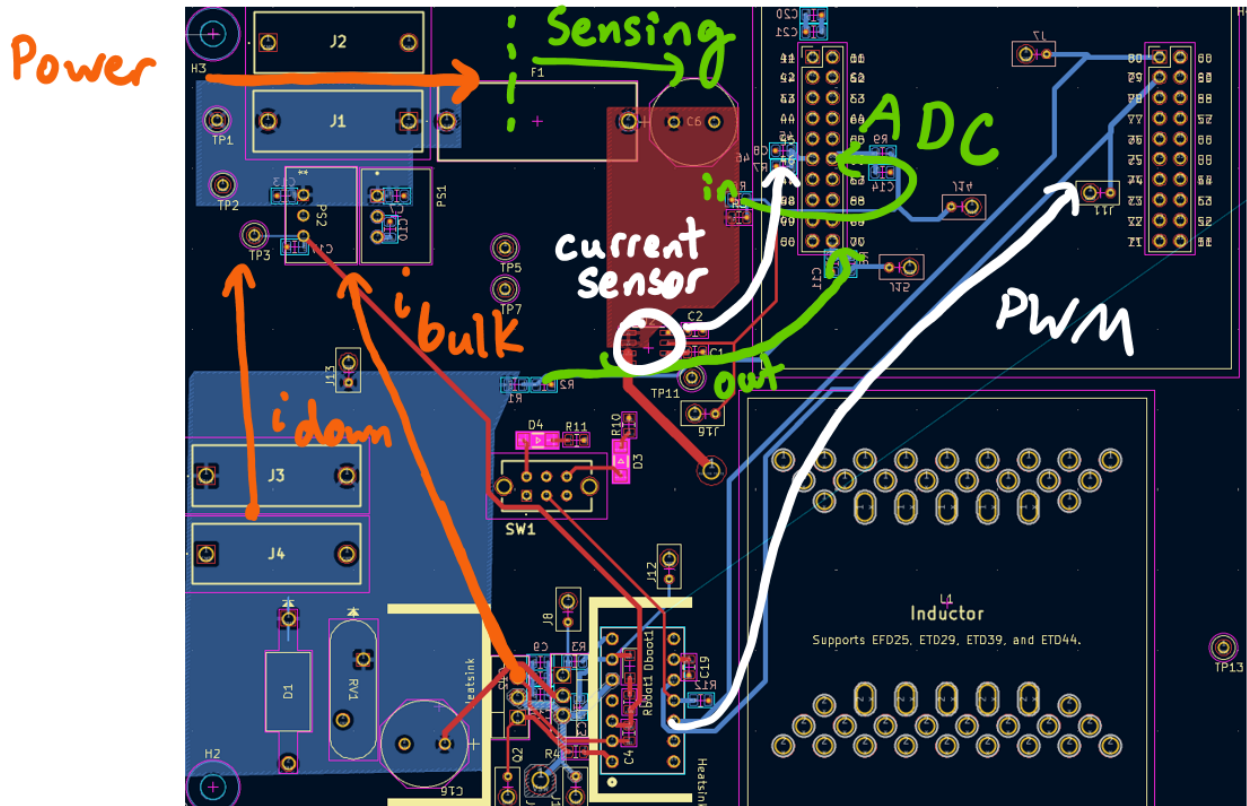


Figure 19: Switch node and nearby layers

3.4 Sensitive Signals

Sensitive ePWM or other sensing signals, like the ADC, should be separated from high $\frac{di}{dt}$ paths. This is due to a strong fluctuating magnetic field generated by high $\frac{di}{dt}$ paths. When a strong magnetic field cuts across the sensitive trace, it can induce unwanted noise (such as voltage or current) into the sensitive signal, negatively affecting signal integrity.

To separate the sensitive signals from high $\frac{di}{dt}$ paths, the C2000 Microcontroller, where all the ePWM signals landed, was placed on right side, while the DC-DC Power Supplies, where high $\frac{di}{dt}$ return paths landed, were placed on the left side of the PCB. As a result, ePWM and ADC signals will not cross over the high $\frac{di}{dt}$ return paths. This can be seen below in Fig. 20.



3.5 3D Layout

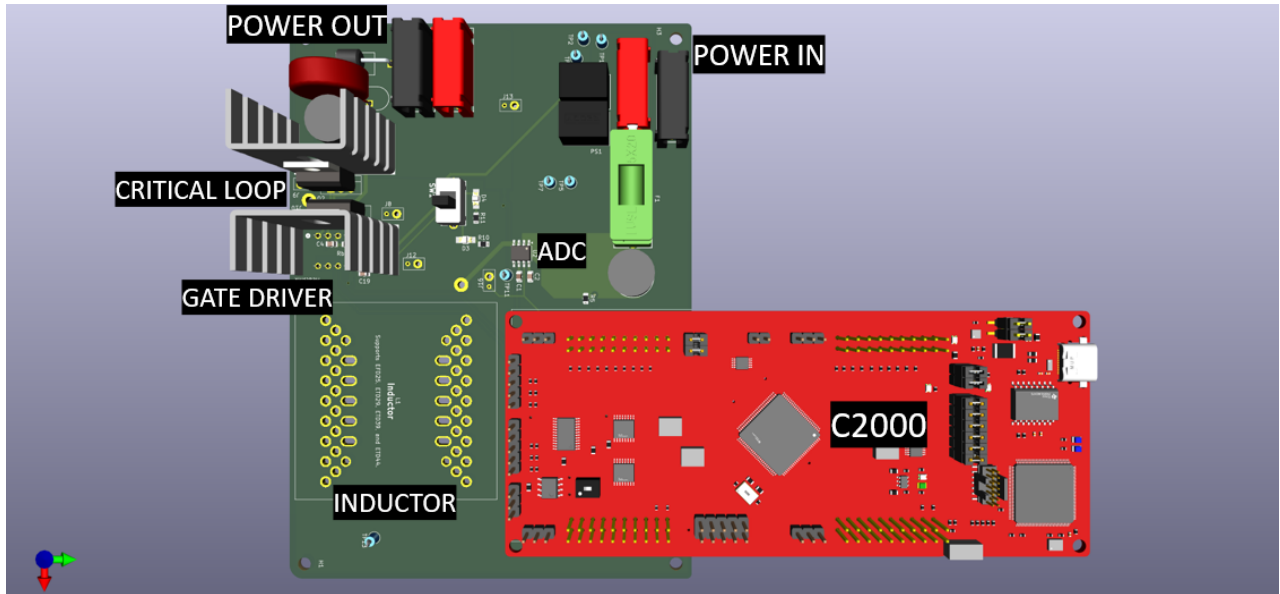


Figure 21: 3D Model of the PCB

3.6 Learning, Success, Struggles

In this lab, we learned that the return path for high $\frac{di}{dt}$ currents cannot be in the vicinity of sensitive signals (e.g., ePWM and ADC signals) to protect their integrity. Therefore, the 24V Terminals were placed on the left, whereas the C2000 were placed on the right. This creates a clean space on the left between the 24V Terminal and the Half-Bridge, providing a clear path for returning high $\frac{di}{dt}$ current while keeping a good distance from the sensing signals on the right. This was one of our original struggles, and it required a complete redesign of the critical loop area to avoid this issue.

Our biggest struggle was keeping the critical loop as compact as possible. We often thought only in 2-D, and even after large improvements from watching the groups before us, we still had to make significant changes after our own design review. Layering the traces on top of each other was a big boon to our critical loop, and so was offsetting the MOSFETs to keep the switch node small.

By the end of the lab, we succeeded in designing a PCB that operates efficiently. By strategically utilizing all four layers and arranging the component layout, the gating loop and the critical loop were minimized to reduce the parasitic inductance generated by the loop, reducing the impedance, hence, reducing the possibility of voltage spikes generated by high $\frac{di}{dt}$ currents. We also had great success in placing our coupling capacitors close to their designated pins from the beginning, reducing the need for re-layout. We also believe that our board is cleanly laid out, which will assist in the soldering and assembly process.

4 Conclusion

4.1 Highs and Lows

Our definite highs so far with the lab were with the overall organization of our PCB. We managed to make great space between our components to avoid noisy signals and allow clean return paths. We are also proud of how tight our loops were by the end of the design process. We also felt great accomplishment from understanding the concepts as we progressed through the labs; for example, in the ADC and PWM labs. It was a great joy to struggle at first and essentially have no understanding of what really was going on, only to slowly progress and reach the eureka moment for each concept.

Our lows were definitely with the critical loop specifically. Our other loops generally did not go through great changes, nor our other components. However, our critical loop went through not just minor changes to the traces, routing, or capacitor placements, but three complete revamps. These involved flipping the entire gate driver to the opposite side of the board, completely rotating our entire critical loop area 90 degrees, and moving the capacitors between the MOSFETs. These changes required a complete rethinking of this area of the board, and almost all traces had to be redone. In the future, we would prepare better by planning ahead for most issues. We started laying out pieces before really understanding, for example, the implications of high $\frac{di}{dt}$ paths.

4.2 Future Goals

Our future goals for the board are to build an overall solid board. Our first priority is efficiency, second is overall cleanliness and aesthetics, and third is weight. While efficiency still takes priority, we want to strike a more even balance than most groups. As we are both new to electrical engineering, we do not want to sacrifice any front too much in our design to maximize our learning of all concepts. We want to have to think about and deal with as many factors as possible, rather than disregarding one for the sake of another. Thus, going forward, we want to select an inductor with the priority of efficiency, but we will not pick the absolute most efficient design if it means sacrificing a significant amount on weight or aesthetics.